Hypothesis of Dynamic Synthesis and Excess Mass Ejection (Słowik Phenomenon)

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This study constitutes an independent theoretical development concerning the physical nature of gravitational collapse and the dynamics of black hole birth. The author introduces a novel view on the mechanism of event horizon formation as a discontinuous, quantized process in the mass parameter space.

1. MAIN ASSUMPTIONS OF THE HYPOTHESIS

It is postulated that the collapse of a massive stellar core above the Tolman-Oppenheimer-Volkoff (TOV) limit does not proceed in a smooth and continuous manner, but behaves like a quantized phase transition. The following pillars of this theory are distinguished:

- Quantization of the starting threshold: The first 'Słowik Package' (1 P = 2.8 M☉) represents an indivisible, minimal portion of matter necessary to break through degenerate pressure barriers and create an event horizon. In nature, there are no intermediate states or 'partially filled' initial packages (e.g., a black hole with a mass of 0.5 P).
- Tendency to energy minimum: During the collapse of a stellar core with a mass greater than the minimum (e.g., 5 M☉), the system tends to form the most stable and compact gravitational state - i.e., a core with a base mass of exactly 2.8 M☉ (1 P).
- Violent rejection of excess: The powerful energy released at the moment of collapse and violent breaking of pressure barriers (generating supernova shock waves and radiation bursts) acts as a repulsive force that violently blows away the outer excess of stellar matter.
- Formation of a buffer (accretion disk): The excess mass of the progenitor that did not fit into the primary collapse core (1 P) is ejected outward, creating a dynamic buffer around the newly formed object in the form of an accretion disk.
- Evolution and smooth growth (N > 1): Only after the formation of a stable core with mass 1 P does the black hole begin slow and continuous accretion of matter from the buffer accumulated in the disk, which allows for gradual, smooth increase of its mass (coefficient N then grows e.g., to 1.1, 1.5 or 2.0).

2. ANALOGY TO NUCLEAR FUSION AND MASS DEFECT

This phenomenon shows a deep and extremely intuitive analogy to processes occurring in nuclear physics during fusion reactions (e.g., formation of helium nuclei from hydrogen):

Process Feature	Nuclear Fusion (Helium Synthesis)	Słowik Phenomenon (Grav. Collapse)
System tendency	Achieving the lowest, most stable energy state of atomic nucleus.	Achieving the most stable, compact gravitational state (1 P = 2.8 M☉).
Reaction to excess	Excess mass cannot remain in new nucleus - released as energy ($E=mc^2$).	Excess mass of star is not immediately absorbed - is ejected by collapse explosion force.
Final effect	Stable quantized alpha particle (helium nucleus) is formed and radiation emitted.	Stable black hole core (1 P) is formed, and buffer (accretion disk) forms around it.

3. ASTROPHYSICAL CONCLUSIONS

The 'Słowik Phenomenon' hypothesis sheds new light on why newly formed stellar black holes show such high consistency in their initial masses. It shows that the accretion disk is not merely a random element of the environment, but a physically required 'safety buffer' that receives excess matter ejected during the quantized phase transition that black hole births are.